

Amandeep Singh

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Work Experience

Adobe

Computer Scientist – II

Jan 2025 – Present

- Building an **ops agent** that executes operational playbooks as skills via an agent loop with **K8s-sandboxed tool execution**.
- Built a **code migration agent** (Claude Code + OpenRewrite) to automate customer AEM codebases to Java 21.
- Representing India team in the Global Architecture Council, contributing to high-level technical decisions.
- Built GCP support for our platform supporting multi-cloud customer initiatives.

Computer Scientist – I

Jan 2023 – Jan 2025

- Built IaC services for **high-security sovereign deployments** in US FedRAMP, Australia IRAP and China.
- Developed the **Content Copy** for Apache Oak repositories, handling **terabytes of daily data transfer** across 20k instances on AWS and Azure.
- Contributed Azure support for upgrade services to modernize **2k+ customer environments** to Red Hat 8.

Member of Technical Staff – II

Aug 2021 – Jan 2023

- Optimized **Blue-Green deployment** workflows, reducing worst-case deployment time by ~1 hour.
- Enhanced **multi-region auto-scaling** by implementing automatic detection of seed instances.
- Mentored and on-boarded a new India-based engineering team as the first member.

99acres

Nov 2019 – Aug 2021

Senior Software Engineer — Android and React Native development for 99acres' app.

Samsung Research Institute

Jun 2018 – Nov 2019

Software Engineer — Android and AOSP Framework development for Samsung's Maximum Power Saving Mode.

Projects

Blackwell NVFP4 Kernels — GPU Mode Competition

Nov 2025 – Feb 2026

- Placed world **#32 in GEMV** and **#36 in GEMM**. Wrote custom **CUDA kernels** for FP4 block-scaled GEMV on B200 — FP4/FP8 decode intrinsics, vectorized loads, warp-shuffle reductions; profiled with **Nsight Compute**.
- GEMM entry used **CUTLASS** and **tcgen05** (Blackwell TMA + Tensor Core) codegen.

Accel — RISC-V ML Accelerator

Mar 2026 – Present

- Building a **hardware ML accelerator** as a Custom Function Unit on a VexRiscv soft-core running on a LiteX SoC, deployed to a **Tang Nano 20K FPGA**.
- **Output-stationary PE array** with int8 SIMD MAC and requantization pipeline in **Amaranth HDL**. Runs models via **Tinygrad** through selective compiler lowering. Bare-metal **Zig firmware**.
- Full simulation via **Renode** + **Verilator** — cycle-accurate CFU co-simulation without hardware.

Steelclaw — Self-hosted AI Agent

Mar 2026

- **Single-binary Rust AI agent**. **Mid-conversation provider failover** across Gemini/Groq/Cerebras/Ollama — sanitizes stateful tool-call history on switch. All state in a **single SQLite database** via AgentFS.

HNSW — High-Performance Vector Search in Zig

Jan 2026 – Mar 2026

- Implemented HNSW from scratch. Benchmarked on **1.3M GloVe vectors: 1950x faster** than brute-force at 75% Recall@10.
- Systematic optimization via **perf**, **cachegrind**, and hardware counters: flat neighbor arrays, f16 vector quantization, generation-counter visited sets, software prefetching, parallel batch construction, and SIMD shuffle reductions — **4x faster build**, **2x faster search** vs initial implementation.

Khojkar — Deep Research Agent

Apr 2025 – May 2025

- **Autonomous research agent from scratch** — custom ReAct loop, tool executor, multi-agent supervisor with shared vector-store memory; no LangChain/CrewAI. Published to **PyPI**.

Education

Dr. B. R. Ambedkar National Institute of Technology, Jalandhar — B.Tech. in ECE

Jul 2014 – Jun 2018

Skills

Python, Rust, Zig, C/CUDA, Java, PyTorch, FastAPI | AWS, Azure, GCP